
RESEARCH PAPER

Society Under Monetary Stress IX: The Fragility Premium

Insurance, Medical Care, Childcare Proxies, and the Cost of Household Resilience

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Abstract

This paper develops the **fragility premium** as a measurement concept: the recurring cost households pay to preserve resilience against shocks that can interrupt work, care, mobility, housing stability, or health. The paper does not claim to estimate a complete household-level premium. Instead, it defines the object formally, separates price growth from exposure and ability-to-pay, and uses public aggregate data to motivate why a household-level index would be useful.

A household-level fragility-premium burden can be written as a resilience-cost bundle divided by household resources: category prices or premiums, multiplied by household exposure or need, divided by income or liquid resources. Public CPI and Consumer Expenditure series do not identify that full object. They can, however, show whether several categories adjacent to household resilience have moved differently from headline prices. From 1998 to 2025, headline CPI rose 97.5 percent. Medical care rose 139.6 percent, medical care services 156.4 percent, hospital services 296.7 percent, motor vehicle insurance 251.2 percent in the available national CPI item, and the tuition/school-fee/childcare aggregate 202.8 percent. Consumer Expenditure health-insurance spending rose 344.1 percent from 1998 to 2024, while healthcare spending rose 225.6 percent. These spending measures are not price indexes.

The revised paper's claim is deliberately narrow: aggregate public data show a pattern consistent with rising prices in several resilience-adjacent categories, and FRED-distributed Consumer Expenditure quintile tables show that healthcare and health-insurance spending absorb larger expenditure shares in the lower-income quintiles. A new Consumer Expenditure public-use microdata appendix extends the measurement exercise to pooled calendar-year 2024 Interview records. The appendix is explicitly a reference-period budget-share exercise, not an annualized household-cost estimate: conservative estimates show mean selected current-quarter resilience costs rising from \$785 below \$25,000 of income to \$2,447 at \$100,000 or more, while the mean household-level cost/income burden falls from 4.8 percent to 1.4 percent. These data still do not prove that all households face a rising fragility burden. That burden is inherently distributional and requires microdata on liquidity, family structure, health risk, care needs, insurance design, geography, transportation exposure, and hardship outcomes. The paper is therefore best read as a public-data motivation and measurement agenda, not as a completed causal household-finance article.

1. Introduction

A household can look stable in aggregate statistics while becoming less resilient in practice. Employment can be positive, nominal income can rise, and credit can remain available, yet the household may still find that the cost of staying protected has increased. Medical exposure, health

insurance, vehicle insurance, renters' or household insurance, commuting reliability, childcare, and out-of-pocket healthcare do not merely purchase comfort. They help keep shocks from becoming crises.

This paper calls that recurring defensive layer the **fragility premium**. The term is useful only if it is treated as a measurement problem, not a slogan. A fragility premium is not the same thing as headline inflation. It is not the same thing as medical inflation. It is not simply the sum of several CPI components. It is the household-specific cost of maintaining access to the protections and services that reduce exposure to disruptive shocks.

The distinction matters because household fragility is not a representative-agent property. A family with young children, one car, a high-deductible health plan, and thin liquid savings faces a different resilience budget than a childless household with remote work, employer-paid benefits, and large liquid buffers. A renter in a high-risk insurance market faces different protection costs than an owner with a different policy structure. A household with chronic illness experiences medical prices and insurance design differently from a household with little expected healthcare use.

The paper's contribution is therefore limited but concrete. First, it defines a household-level fragility-premium object that separates prices, exposure, coverage, utilization, and ability to pay. Second, it shows that several public aggregate proxies near this margin have risen faster than headline CPI. Third, it adds a descriptive CEX public-use microdata appendix showing selected reference-period resilience-cost budget shares by income group in pooled calendar-year 2024 Interview records. Fourth, it explains why these patterns are still not enough: a serious fragility-premium account must connect household-level burdens to liquidity and hardship outcomes.

This repositioning is intentional. The paper is not offered as a causal design or completed household-level measurement article. It is a descriptive research note and measurement agenda. Its public-data exercise is useful as motivation, but the central empirical claim remains bounded: selected resilience-adjacent prices and spending measures have grown faster than broad consumer prices, making it plausible that some households face a rising cost of staying resilient.

2. Defining the fragility premium

A simple way to organize the concept is:

$$FPh_t = [sum_k ehkt \ pkt \ qhkt + ohkt] / R_{ht}$$

where h indexes households, t time, and k resilience-related categories. *pkt* is the relevant price, premium, or service cost; *qhkt* is quantity or utilization; *ehkt* is household exposure or need; *ohkt* is out-of-pocket residual exposure not captured by posted prices or premiums; and *R_{ht}* is the household resource denominator, such as income, liquid assets, or cash flow after fixed obligations.

This accounting identity clarifies four points.

First, a price index is not the full premium. A hospital-services CPI index can rise sharply, but household burden depends on probability of use, insurance coverage, deductibles, coinsurance, provider networks, income loss, and liquidity. Second, spending is not a price index. Consumer Expenditure healthcare spending can rise because of prices, utilization, coverage changes, demographics, or reporting composition. Third, a category can matter more for some households than others. Childcare prices are most relevant for families with young children and caregivers whose labor supply depends on care arrangements. Auto insurance matters more for car-dependent households. Fourth, the same price shock is more or less burdensome depending on R_{ht} : income, liquid savings, debt service, and access to emergency liquidity.

This formalization turns the paper's intuition into a testable measurement agenda. A complete fragility-premium paper would estimate FP_{ht} or a close approximation using household-level data. The current public-data note cannot do that. It can only examine the price and spending components that are visible in aggregate series.

3. Relation to existing research

The idea sits near several established literatures. Work on household financial fragility emphasizes liquidity and emergency resources. Lusardi, Schneider, and Tufano study households' ability to raise emergency funds, while Kaplan, Violante, and Weidner show that households can be "wealthy hand-to-mouth" when illiquid wealth coexists with low liquid buffers. That literature explains why shocks can bind even when net worth or income looks adequate.

Medical-shock research provides a direct mechanism for the fragility-premium idea. Dobkin et al. study the economic consequences of hospital admissions and show that health shocks affect out-of-pocket spending, unpaid bills, bankruptcy, earnings, income, and credit access. Hospital-price and provider-market research, including work by Cooper, Craig, Gaynor, and Van Reenen, shows that medical prices vary substantially and can affect spending for privately insured households. This paper's hospital-services evidence should be read against that background: rising hospital prices matter for fragility because medical shocks can transmit through both expenses and labor income.

Childcare and labor-supply research also bears directly on the concept. Childcare costs affect whether parents, especially mothers, can maintain labor-force attachment. A childcare price series is therefore not only a consumption price; for some households it is a work-continuity cost. The present draft uses only a broad tuition/school-fee/childcare aggregate because that is the reproducible public series included in the current bundle. A stronger version should replace that proxy with narrower day-care/preschool and childcare-price data, and should connect the burden to labor-supply outcomes.

Finally, CPI and insurance-measurement literatures matter because several candidate components are hard to interpret. Health insurance CPI is not a simple household premium series. Its retained-earnings methodology and post-2023 methodological changes make it especially unsuitable as a

standalone measure of household premium burden. The revised paper therefore treats health insurance CPI as a cautionary public series, while relying more on the broader claim that the household-level object requires better microdata.

4. Data and measurement boundaries

The public-data exercise uses FRED and BLS public series. The main window is 1998-2025 where available. Some Consumer Expenditure rows end in 2024. Monthly series are converted to annual averages. Growth tables compare endpoint changes and compound annual growth rates. Episode tables show annual average year-over-year changes for selected years. Differential tables report annual inflation differentials relative to headline CPI where observations are available. The draft also uses FRED-distributed Consumer Expenditure quintile tables for total expenditures, healthcare spending, and health-insurance spending in 2024; these are aggregate tabulations of complete income reporters, not household-level microdata.

The main CPI price series are headline CPI (CPIAUCSL), medical care (CPIMEDSL), medical care services (CUSR0000SAM2), hospital and related services (CUSR0000SEMD), tenants' and household insurance (CUUR0000SEHD), health insurance (CUUR0000SEME), motor vehicle insurance (CUUR0000SETE), and tuition, other school fees, and childcare (CUSR0000SEEB). The denominator series are real disposable personal income (DSPIC96), disposable personal income (DSPI), average hourly earnings of production and nonsupervisory employees (AHETPI), and median household income (MEH0INUSA646N). Consumer Expenditure healthcare and health-insurance spending (CXUHEALTHLB0101M, CXUHLTHINSRLB0101M) are included as spending measures, not price indexes. The quintile rows use CXUT0TALEXPLB01xxM, CXUHEALTHLB01xxM, and CXUHLTHINSRLB01xxM, where xx indexes the lowest through highest income quintiles.

A compact classification helps keep the measurement boundaries clear:

Variable family	Examples	Role in this note	Main limitation
Price indexes	CPI medical care, hospital services, motor vehicle insurance, tenants' and household insurance	Show category price pressure relative to headline CPI	Do not measure household exposure, utilization, insurance design, or liquidity
Spending measures	CE healthcare, health-insurance spending, CEX selected resilience costs	Show reported household or grouped expenditure burdens	Spending can be low because need is low, because coverage is generous, or because households forgo care/services
Denominators	Income, total expenditures, wages, disposable income	Scale costs by household or aggregate resources	Annual income and liquid resources are not interchangeable
Exposure proxies	Childcare/care categories, vehicle insurance, housing insurance	Indicate margins where household needs differ	Current public-data version cannot fully observe need or risk exposure
Outcomes	Delayed care, unpaid bills, liquidity stress, labor supply	Needed for a causal hardship paper	Not estimated in this descriptive note

The boundaries are central, not peripheral. Health insurance CPI is methodologically unusual and should not be read as direct household premium inflation. Consumer Expenditure health-insurance spending is nominal spending and reflects prices, utilization, coverage, employer benefit arrangements, household composition, and reporting. The broad tuition/school-fee/childcare CPI series is not a pure childcare index. National aggregate indexes also hide variation by income, family structure, region, health status, insurance design, housing tenure, and vehicle dependence.

For that reason, the tables below should not be interpreted as measuring *FPh_t*. *They describe public price and spending series that plausibly enter a future FPh_t measure.* The difference is important.

5. Public-data motivation: long-run growth

Table 1 gives the basic descriptive motivation. Headline CPI rose 97.5 percent from 1998 to 2025. Several resilience-adjacent categories rose faster: medical care, medical services, hospital services, motor vehicle insurance, and the broad tuition/school-fee/childcare aggregate. Consumer Expenditure healthcare and health-insurance spending rose even more, but those rows are spending measures and should not be compared mechanically to CPI item indexes.

Table 1. Growth in selected household resilience-cost proxies. Growth is measured over each series' available years within 1998-2025. CPI item indexes measure prices, while Consumer Expenditure rows measure nominal annual expenditures and are not directly comparable price indexes.

Series	ID	Start	End	Series growth %	Series CAGR %	Headline CPI growth %	Headline CAGR %
Headline CPI	CPIAUCSL	1998	2025	97.5	2.55	97.5	2.55
Medical care	CPIMEDSL	1998	2025	139.6	3.29	97.5	2.55
Medical care services	CUSR0000SAM2	1998	2025	156.4	3.55	97.5	2.55
Hospital services	CUSR0000SEMD	1998	2025	296.7	5.24	97.5	2.55
Health insurance CPI	CUUR0000SEME	2005	2025	47.1	1.95	64.9	2.53
Health insurance CE spending	CXUHLTHINSRLB0101M	1998	2024	344.1	5.90	92.4	2.55
Healthcare CE spending	CXUHEALTHLB0101M	1998	2024	225.6	4.65	92.4	2.55
Motor vehicle insurance	CUUR0000SETE	1998	2025	251.2	4.76	97.5	2.55
Tenants' & household insurance	CUUR0000SEHD	1998	2025	66.1	1.90	97.5	2.55
Tuition/school fees/childcare	CUSR0000SEEB	1998	2025	202.8	4.19	97.5	2.55
Real disposable personal income	DSPIC96	1998	2025	95.1	2.51	97.5	2.55
Average hourly earnings	AHETPI	1998	2025	141.0	3.31	97.5	2.55
Median household income	MEHOINUSA646N	1998	2024	115.3	2.99	92.4	2.55

Figure 1 shows medical-price categories indexed to 1998. Hospital services stand out. The relevant household mechanism is not that every household buys hospital services each year. It is that hospital exposure is a low-frequency, high-consequence event whose financial effects depend on insurance design, income loss, liquidity, and billing risk. Public CPI data show price pressure in the category; they do not measure household exposure to that pressure.

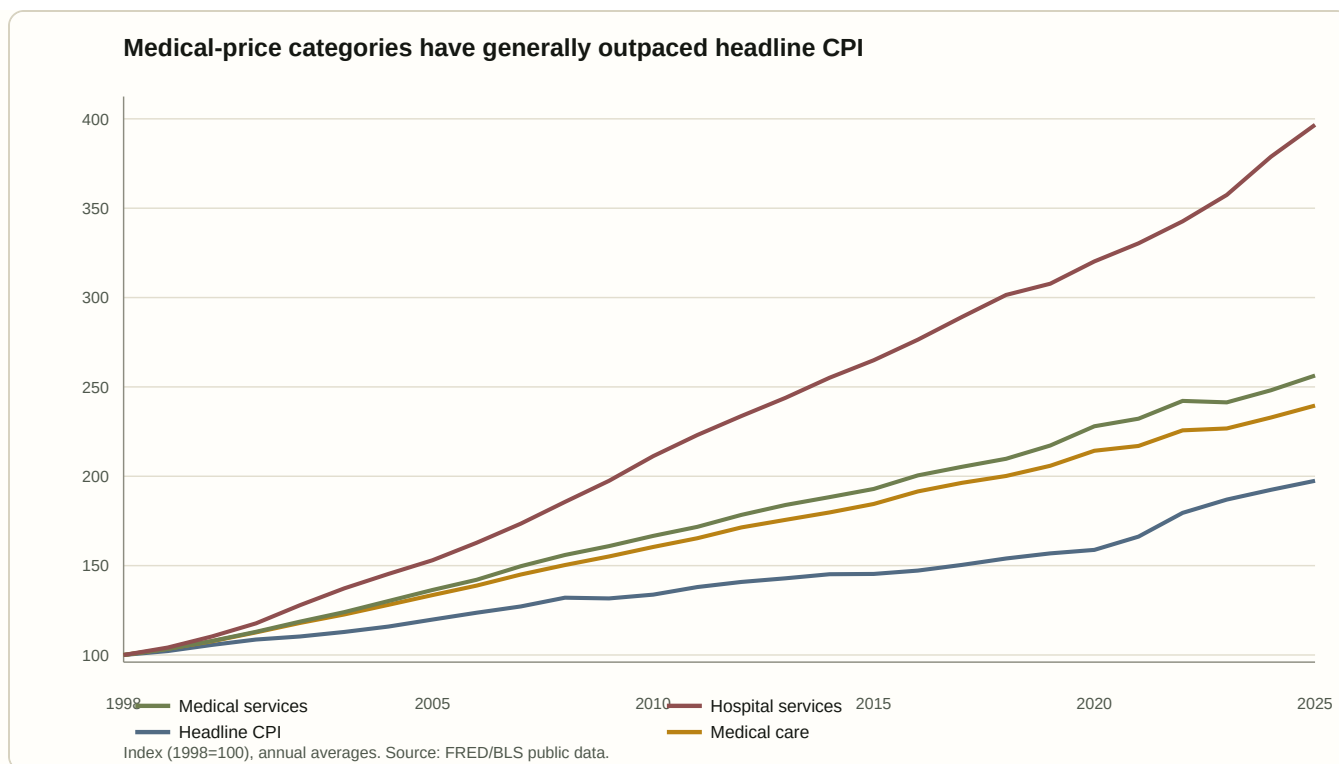


Figure 1. Medical-price pressure versus headline CPI.

Figure 2 shows insurance-related categories. The health insurance CPI line should be read cautiously for the reasons above. Motor vehicle insurance, however, is a clearer example of a household-facing risk-management cost that rose sharply after 2020. Even there, aggregate price movement is not the full household burden: car dependence, driving record, location, vehicle type, and state insurance regulation all matter.

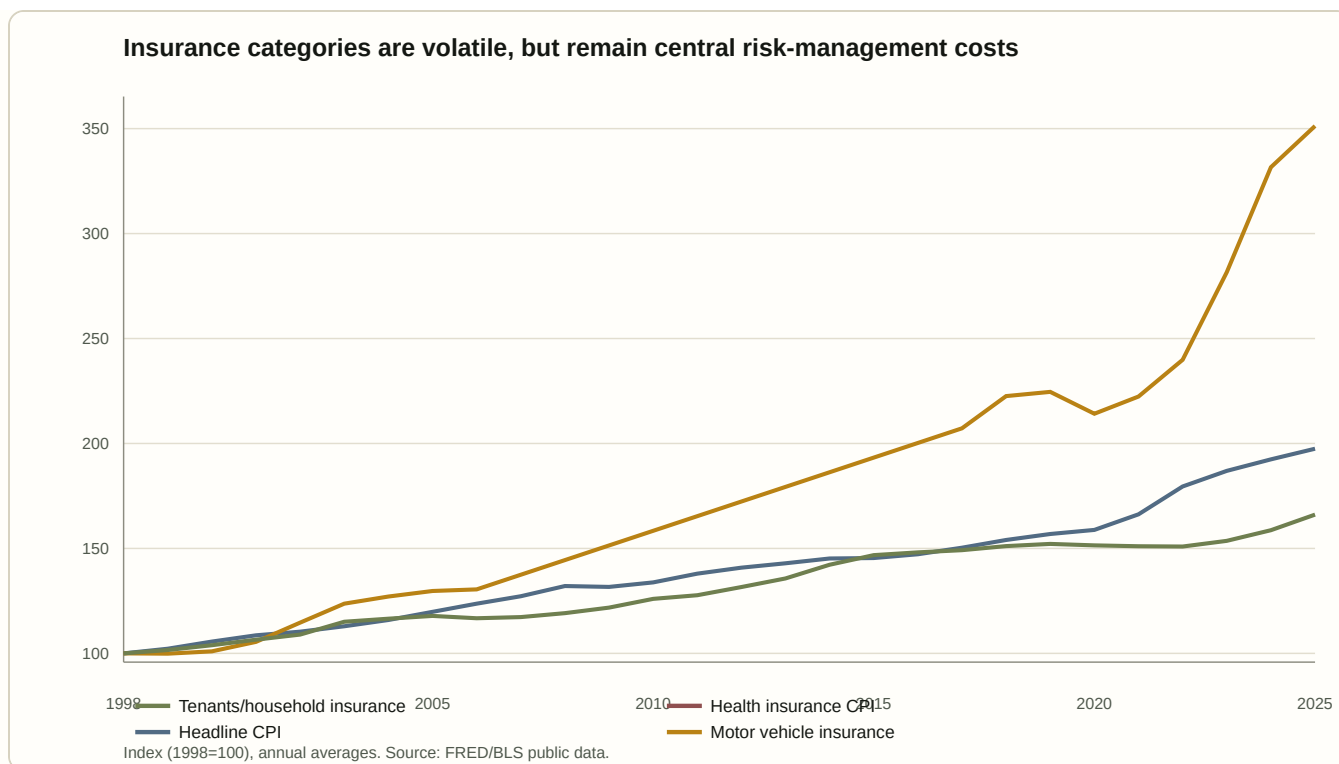


Figure 2. Insurance price pressure versus headline CPI.

Figure 3 compares the broad tuition/school-fee/childcare aggregate with headline CPI and average hourly earnings. The proxy is too broad for strong childcare claims. It is included because education and care-related fees sit near the family-formation and work-continuity margin, but a future version should replace or supplement it with narrower childcare price and expenditure data.

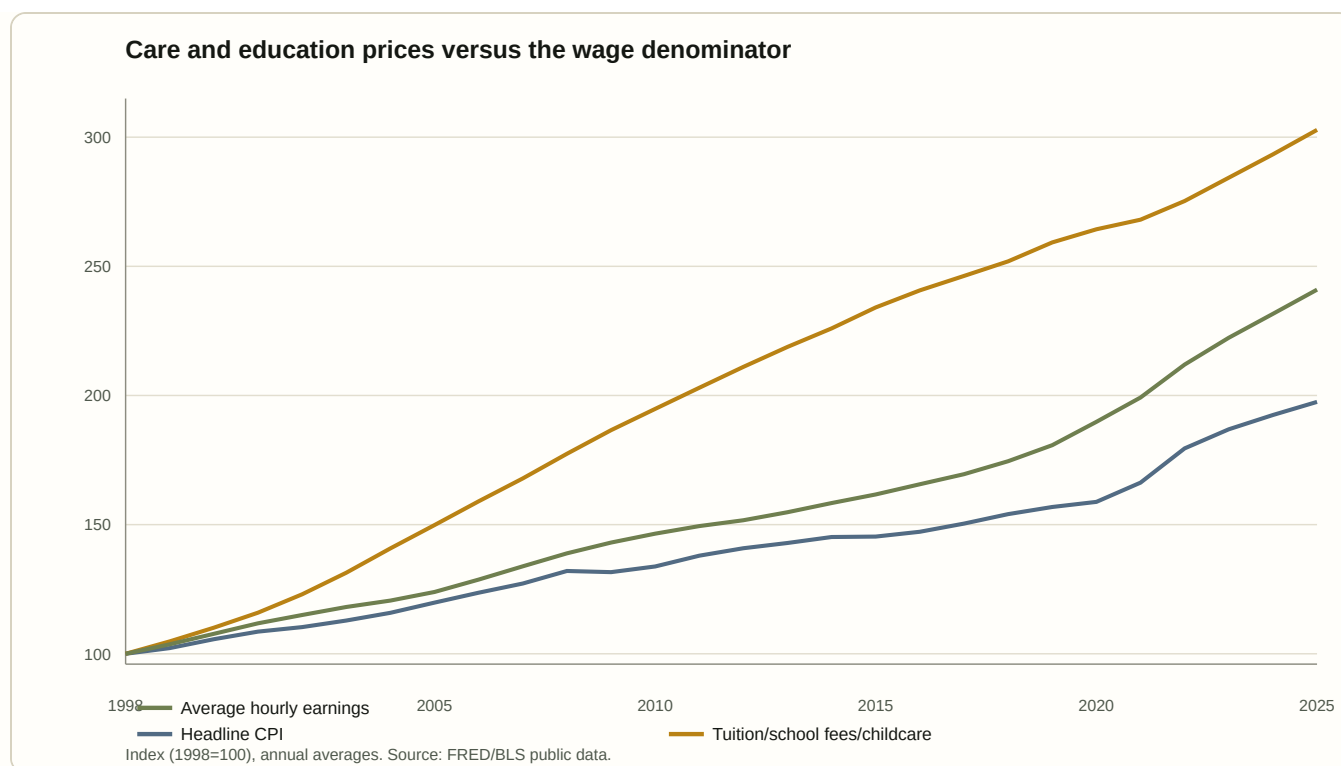


Figure 3. Care and education prices versus the wage denominator.

6. Episode evidence

Table 2 shows selected years. The table is useful mainly because it shows timing. Some resilience-adjacent pressures appear during periods when headline inflation is not the whole story. In 2024, for example, headline CPI growth cooled to 3.0 percent while motor vehicle insurance rose 17.8 percent and hospital services rose 6.0 percent. A household that had already absorbed the 2021-2022 inflation shock could still face delayed insurance and medical-cost adjustments.

Table 2. Episode-year inflation and income context. Entries are annual average year-over-year percent changes. Real disposable personal income is shown as an income-side denominator, not a price category.

Episode	Year	Headline	Medical	Medical services	Hospital	Health ins.	Auto ins.	Household ins.	Tuition/childcare	Real DPI
2001 recession	2001	2.8	4.6	4.8	6.6	—	4.4	2.5	5.3	2.9
financial crisis	2008	3.8	3.7	4.2	7.0	—	—	1.6	5.7	1.6
pre-pandemic baseline	2019	1.8	2.8	3.5	2.1	14.5	0.9	0.7	2.9	3.1
inflation shock	2022	8.0	4.1	4.3	3.7	14.0	7.9	-0.1	2.7	-5.7
post-shock adjustment	2024	3.0	2.7	2.8	6.0	-5.6	17.8	3.3	3.2	2.9
latest full year	2025	2.6	2.9	3.3	4.7	3.0	5.9	4.7	3.2	1.7

7. Relative inflation differentials

Table 3 reports annual differentials relative to headline CPI. Hospital and related services averaged 2.6 percentage points above headline inflation in the 1998-2025 window. The broad tuition/school-fee/childcare aggregate averaged 1.7 percentage points above headline. Medical services averaged 1.0 point above headline and medical care 0.8 point above. Motor vehicle insurance averaged 2.0 points above headline where observations are available, with a median near zero because the series combines quiet periods with sharp jumps.

Table 3. Annual inflation differentials relative to headline CPI, 1998-2025 where available. Positive values mean the category rose faster than headline CPI in that year.

Series	ID	N	Mean pp above headline	Median pp	Max pp	Min pp
Medical care CPI	CPIMEDSL	28	0.8	1.0	3.5	-3.9
Medical care services CPI	CUSR0000SAM2	28	1.0	1.5	3.7	-4.5
Hospital and related services CPI	CUSR0000SEMD	28	2.6	2.8	7.1	-4.3
Health insurance CPI	CUUR0000SEME	9	-1.4	-0.1	14.6	-26.9
Motor vehicle insurance CPI	CUUR0000SETE	17	2.0	-0.1	14.8	-5.9
Tenants' and household insurance CPI	CUUR0000SEHD	28	-0.7	-1.1	3.3	-8.1
Tuition, other school fees, and childcare CPI	CUSR0000SEEB	28	1.7	1.9	5.5	-5.3

The distributional implication cannot be inferred from this table alone. A category with a modest average differential may still produce large household stress if it spikes when buffers are thin or if it applies to households with high exposure. Conversely, a category can outpace headline CPI without materially affecting a household that does not use it. This is why the aggregate exercise is a motivation, not the final measurement.

8. Public distributional tabulation: health burden by income quintile

Table 4 adds a limited distributional check using FRED-distributed Consumer Expenditure quintile tables. This is not the microdata upgrade that a publishable household-level paper would require, but it is useful because it moves one step beyond national aggregate averages. In 2024, healthcare spending absorbed 9.8 percent of total expenditures in the lowest income quintile and 6.5 percent in the highest income quintile. Health-insurance spending absorbed 6.1 percent of total expenditures in the lowest quintile and 4.1 percent in the highest quintile. In dollar terms, higher-income households spent more, but as a share of the consumption budget the burden was about 1.5 times larger in the lowest quintile than in the highest quintile.

Table 4. Consumer Expenditure healthcare and health-insurance spending by income quintile, 2024. Expenditure levels are annual dollars for complete income reporters. Shares divide category spending by total annual expenditures within each quintile. These are spending burdens, not price indexes.

Income group	Total exp.	Healthcare exp.	Healthcare share %	Health-ins. exp.	Health-ins. share %
Lowest 20%	\$35,046	\$3,445	9.8	\$2,153	6.1
Second 20%	\$50,054	\$4,826	9.6	\$3,194	6.4
Middle 20%	\$66,900	\$5,676	8.5	\$3,842	5.7
Fourth 20%	\$89,972	\$7,247	8.1	\$4,913	5.5
Highest 20%	\$150,342	\$9,771	6.5	\$6,160	4.1
Lowest/highest ratio	0.23	0.35	1.51	0.35	1.50

The table strengthens the measurement point without changing the paper's evidentiary status. It shows that even a public grouped tabulation contains the asymmetry that the fragility-premium concept predicts: resilience-related spending can be lower in dollars for lower-income households yet higher as a budget share. But it still cannot identify within-quintile liquidity, health status, insurance design, delayed care, geography, household size, or exposure to medical shocks. Those are precisely the margins where microdata are needed.

9. CEX microdata appendix: CY2024 resilience-cost budget shares

A final hardening step adds a Consumer Expenditure Survey public-use microdata appendix. The appendix should be read as a descriptive measurement extension, not as a causal hardship design. It uses official 2023 and 2024 Interview public-use files to create a pooled calendar-year 2024 Interview sample: quarter 1 from the 2023 package and quarters 2 through 4 from the 2024 package. Duplicate 241x files in the 2024 package are excluded. The preferred conservative map includes healthcare out-of-pocket spending, health insurance payments, vehicle insurance, homeowners or tenant insurance, and a childcare summary fallback. It excludes second-home homeowners insurance and convalescent or nursing-home care from the main table because those are better handled as sensitivities.

The CEX appendix is useful because it observes consumer-unit spending and denominators rather than only aggregate prices or published quintile tables. The table is not annualized. Dollar values are weighted means of selected current-quarter/reference-period reported costs for pooled 2024 Interview records. The ratio columns are weighted means of consumer-unit-level ratios, not ratios of aggregate weighted means. The income-burden column is computed only for consumer units with at least \$10,000 in income to reduce denominator artifacts. The expenditure-burden column uses positive total-expenditure denominators. Because it is a mean of household ratios, it need not equal mean selected cost divided by mean total expenditure.

In dollar terms, selected current-quarter resilience costs rise with income. As a share of income, however, the burden falls with income. The conservative estimates show mean selected costs of \$785 for consumer units below \$25,000 in income and \$2,447 for consumer units at or above \$100,000. As a share of income among consumer units with at least \$10,000 in income, the corresponding burdens are 4.8 percent and 1.4 percent.

Table 5. CEX CY2024 selected resilience-cost burden by income group. Estimates use official BLS Consumer Expenditure Interview public-use microdata. Calendar-year 2024 combines Q1 from int rvw23 with Q2-Q4 from int rvw24, excluding duplicate 241x files. N CUs is the unweighted consumer-unit count; weighted CUs are average quarterly represented consumer units, computed as the sum of final weights across selected quarters divided by four. The conservative map includes healthcare out-of-pocket costs, health insurance, vehicle insurance, homeowners or tenant insurance, and a preliminary childcare summary fallback. Standard errors in parentheses use the 44 public-use replicate weights and the BLS balanced repeated replication convention, computed as the square root of the average squared difference between replicate and full-sample estimates. Dollar values are current-quarter/reference-period costs, not annualized household costs. Ratio columns report weighted means of consumer-unit-level ratios. The table is descriptive and should not be interpreted as causal hardship evidence.

Income group	N CUs	Weighted CUs	Positive-cost share	Mean cost (SE)	Cost / income (SE)	Cost / total expenditure (SE)
< \$25k	2,952	21.7m	73.9%	\$785 (\$31)	4.8% (0.2%)	19.2% (0.6%)
\$25k-\$50k	3,643	26.0m	83.3%	\$1,270 (\$37)	3.5% (0.1%)	21.4% (0.6%)
\$50k-\$100k	4,844	35.9m	86.9%	\$1,584 (\$33)	2.2% (0.0%)	18.9% (0.5%)
>= \$100k	7,090	51.3m	89.3%	\$2,447 (\$46)	1.4% (0.0%)	16.7% (0.4%)

The appendix does not eliminate the paper's main limitation. CEX still measures spending, not unmet need. A household can have low medical spending because it is healthy, because it is well insured, or because it delays care. Childcare also requires care because the detailed childcare UCCs are not observed in the 2024 MTBI rows used here; the table therefore uses the FMLI childcare summary variable as a preliminary fallback. The appendix strengthens the measurement case, but MEPS and SHED remain better sources for access barriers, medical financial strain, emergency-buffer fragility, and hardship outcomes.

10. What the aggregate evidence can and cannot support

The evidence supports a narrow descriptive statement: several public categories adjacent to household resilience have risen faster than headline CPI over the last generation. That statement is useful because headline inflation alone can understate stress in categories that keep households medically covered, mobile, insured, and able to work.

The evidence does not prove that all households face a rising fragility premium. Table 4 estimates grouped spending shares for healthcare and health insurance, but the broader evidence still does not estimate full household fragility-premium burdens, liquid-resource pressure, insurance take-

up, delayed care, borrowing, arrears, labor-supply changes, or default. It does not separate price from utilization or coverage. It does not identify causal mechanisms. It also does not establish that the broad childcare proxy measures childcare burden.

The strongest conclusion is therefore conditional: if a household has high exposure to medical, care, vehicle, or insurance categories, and if its income or liquid resources do not keep pace, then faster growth in those categories can raise the cost of household resilience. Public aggregate data show why that condition is plausible. They do not show who bears it, how much, or with what behavioral consequences.

11. What a publishable household-level paper would require

A stronger version would move from aggregate price proxies to household-level burden measures. It would estimate something close to `FP_ht` using Consumer Expenditure microdata for spending shares, Medical Expenditure Panel Survey data for medical utilization and out-of-pocket exposure, childcare-price data for families with young children, and insurance-premium or rate-filing data where available.

The key tables would be distributional. They should report resilience-cost burden by income quintile, liquid-asset position, parental status, renter/owner status, age, region, health-risk status, insurance coverage, and vehicle dependence. They should decompose changes into prices, utilization/exposure, coverage, and income denominators.

The next step would connect the burden to outcomes: liquid savings, debt service, revolving credit, unpaid bills, delayed medical care, reduced childcare usage, reduced labor supply, family formation, or reported financial stress. A causal design might exploit Medicaid expansion, childcare price variation, auto-insurance regulatory variation, hospital-market concentration, or insurance-rate shocks. Without one of these extensions, the present paper should remain a descriptive public-data note.

12. Conclusion

The fragility premium is best understood as a household-level resilience burden: the cost of maintaining protections against shocks, divided by the resources available to absorb those costs. Public aggregate data cannot measure that burden directly. They can, however, show why the burden deserves attention.

The public evidence in this note points to sustained price or spending pressure in several categories near household resilience: hospital services, medical services, healthcare spending, motor vehicle insurance, and broad care/education fees. The added quintile tabulation and CEX public-use appendix show that selected healthcare, insurance, and resilience-cost spending also

take larger resource shares in lower-income groups. These patterns do not prove household fragility. They do show that the cost of staying protected is not always well represented by headline CPI or aggregate averages.

The practical lesson is disciplined rather than sweeping. Do not infer household resilience from employment, income, or headline inflation alone. Resilience also depends on whether households can afford the protections that keep shocks from becoming crises. Measuring that properly requires microdata. This paper defines the object and provides the public-data motivation for building it.

Data appendix and replication note

All source series used in this draft are public. FRED series are downloaded through the FRED graph CSV endpoint. BLS item series used here are retrieved through the BLS public API where available. The replication workflow rebuilds the annual panel, processed tables, and figures from those public source series.

Annual values are annual averages of monthly source values unless the source is already annual. Growth rates are endpoint changes over the available years inside the 1998-2025 window. Consumer Expenditure rows ending in 2024 are not extended to 2025. Missing observations are not interpolated. Category differentials are annual year-over-year category inflation minus annual year-over-year headline CPI inflation. Quintile expenditure shares divide each quintile's healthcare or health-insurance spending by that quintile's total annual expenditures in the same year. The CEX microdata appendix is produced by `scripts/buildcexresiliencesmokeable.py`; pooled calendar-year 2024 outputs are written under `data/cex/work/cexresilienceburdency2024*`, with readable tables under `tables/cex/`. The CEX dollar estimates in Table 5 are current-quarter/reference-period amounts from pooled Interview records, not annualized household totals; ratio columns are weighted means of consumer-unit-level ratios.

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